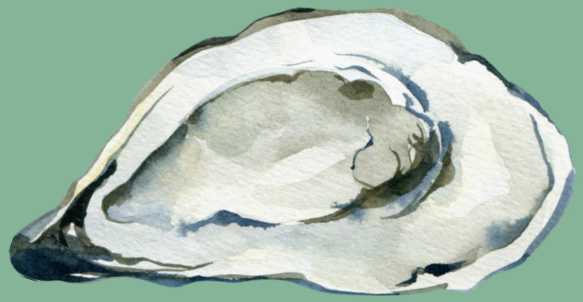
Eastern Oyster

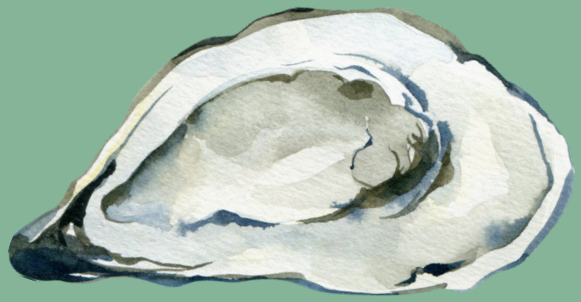
(*Crassostrea virgínica*)



Survival Kit

Ostrea

Eastern Oyster
(*Crassostrea virginica*)



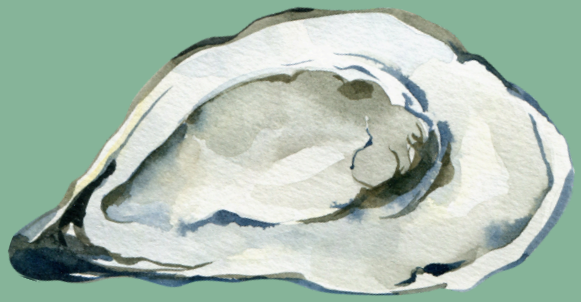
Survival Kit

- Food

Food

Oysters are filter feeders, which means they get their food by filtering tiny particles out of the water. Oysters take in water and capture these tiny food particles as the water passes through their bodies.

Eastern Oyster
(*Crassostrea virginica*)



Survival Kit

- Food
- Dissolved Oxygen

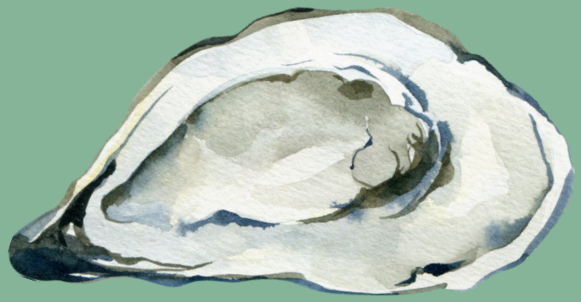
Dissolved Oxygen

Instead of breathing oxygen from the air, oysters take in dissolved oxygen, which is oxygen mixed into the water. Water moves through the oyster's body, allowing it to take in the oxygen it needs.

Hypoxia: A condition where the amount of dissolved oxygen in water is unusually low.

Decomposition: The process of dead plants and animals breaking down. Decomposition can use oxygen in the water and cause oxygen levels to decrease.

Eastern Oyster
(*Crassostrea virginica*)



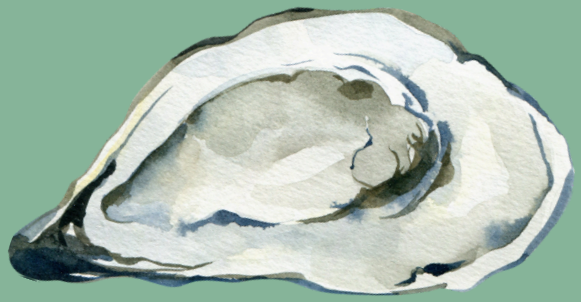
Survival Kit

- Food
- Dissolved Oxygen
- Salinity

Salinity

Salinity is the amount of dissolved salt in water. Oysters in the Chesapeake Bay live in brackish water, which contains a mixture of freshwater and saltwater. Salinity can change with changes in the amount of freshwater entering the Bay. Oysters can survive within a range of salinities, but very large or sudden changes can stress them.

Eastern Oyster
(*Crassostrea virginica*)



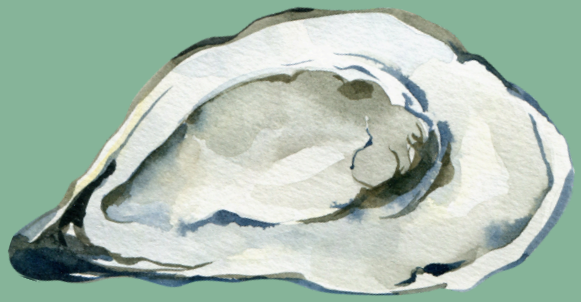
Survival Kit

- Food
- Dissolved Oxygen
- Salinity
- Water Temperature

Water Temperature

Oysters are ectotherms, which means their body temperature is strongly affected by the temperature around them. Oysters can experience seasonal temperature changes, but extremely hot or cold conditions can cause stress.

Eastern Oyster
(*Crassostrea virginica*)



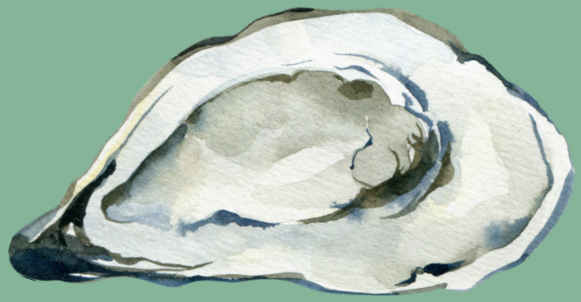
Survival Kit

- Food
- Dissolved Oxygen
- Salinity
- Water Temperature
- Water Quality

Water Quality

Oysters need water that is healthy enough for them to feed, breathe, grow, and reproduce. Water quality can be affected by pollution, excess nutrients, sediment, chemicals, and other materials entering the water. Poor water quality can make it harder for oysters to feed and can damage the habitats where they live. Healthy water does not have to be perfectly clear, but it needs to have conditions that support living organisms.

Eastern Oyster
(*Crassostrea virginica*)



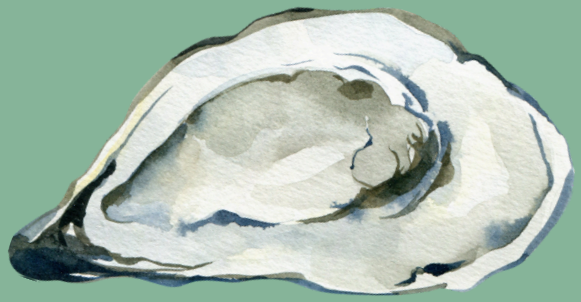
Survival Kit

- Food
- Dissolved Oxygen
- Salinity
- Water Temperature
- Water Quality
- Sediment and Turbidity

Sediment and Turbidity

Sediment is material such as soil, sand, clay, and other small particles that can be carried into the water. Heavy rain and runoff can move large amounts of sediment from land into rivers, streams, and the Chesapeake Bay. Too much sediment can make the water cloudy and can cover oyster shells and other surfaces where oysters live. Very cloudy water can also make it harder for oysters to feed normally.

Eastern Oyster
(*Crassostrea virginica*)



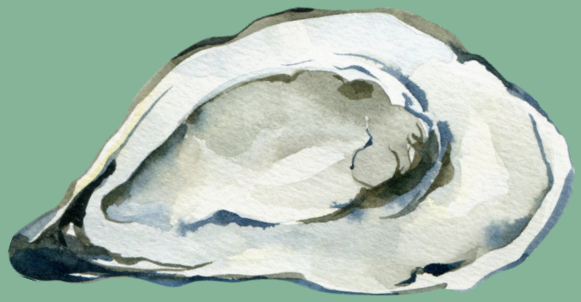
Survival Kit

- Food
- Dissolved Oxygen
- Salinity
- Water Temperature
- Water Quality
- Sediment and Turbidity
- Hard Surfaces

Hard Surfaces

Young oysters need a hard surface where they can attach and continue growing. Oyster larvae drift through the water until they find a suitable surface, such as an old oyster shell or an oyster reef. Once a larva attaches, it becomes a young oyster called spat and begins growing its shell. Having enough hard surfaces is important because they give new oysters places to settle and grow.

Eastern Oyster
(*Crassostrea virginica*)



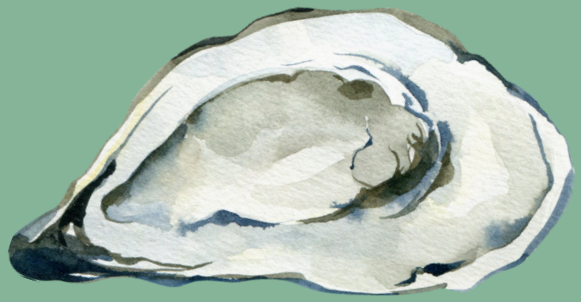
Survival Kit

- Food
- Dissolved Oxygen
- Salinity
- Water Temperature
- Water Quality
- Sediment and Turbidity
- Hard Surfaces
- Habitat

Habitat

A habitat is the place where an organism lives and gets what it needs to survive. An oyster's habitat includes not only the water around it, but also things like food, oxygen, temperature, salinity, and surfaces where oysters can attach. Oyster reefs can also create habitat for fish, crabs, worms, and other organisms. This means oysters both depend on habitat and help create habitat.

Eastern Oyster
(*Crassostrea virginica*)



Survival Kit

- Food
- Dissolved Oxygen
- Salinity
- Water Temperature
- Water Quality
- Sediment and Turbidity
- Hard Surfaces
- Habitat
- Nutrients/Algal Bloom

Nutrients/Algal Bloom

Nutrients are important because organisms need them to grow, but too many nutrients can cause problems in an aquatic ecosystem. Extra nutrients can cause algae and phytoplankton to grow very quickly, creating an algal bloom. When large amounts of algae die, decomposers break them down and can use up oxygen in the water. This can lower dissolved oxygen and create stressful conditions for oysters and other animals.